Practice Intermediate Algebra Test II (version 2)

Name
Date
Course

I) Write your name, the date, and the course title (i.e. Intermediate Algebra).
II) Show all work.
III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
IV) Read carefully. Make sure that you answer only what each question asks.
V Each problem is worth four points for a total of 100 points.

① Convert the equation below to vertex form.

$$y=(x+3)(x+7)$$

② Convert the equation below to intercept form.

$$3(x+1)^2-12=y$$

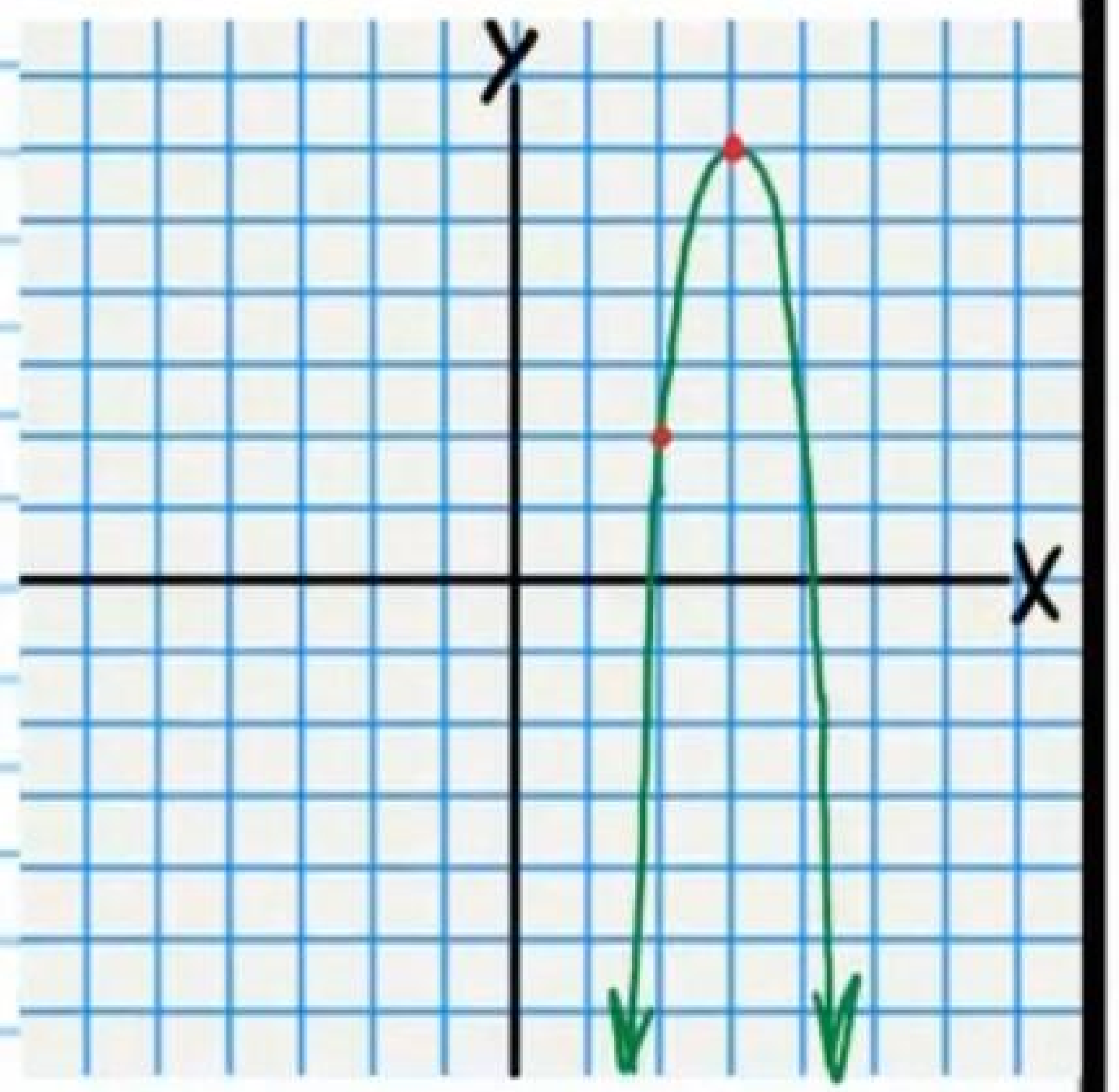
③ Convert the equation below to the two other forms (i.e. intercept and vertex).

$$5x^2+30x+25=y$$

④ Find the discriminant of the quadratic expression below and use it to determine the number of x-intercepts of the corresponding graph.

$$-x^2+x+1=y$$

⑤ Write an equation of the parabola. Hint: It is easiest to use vertex form.



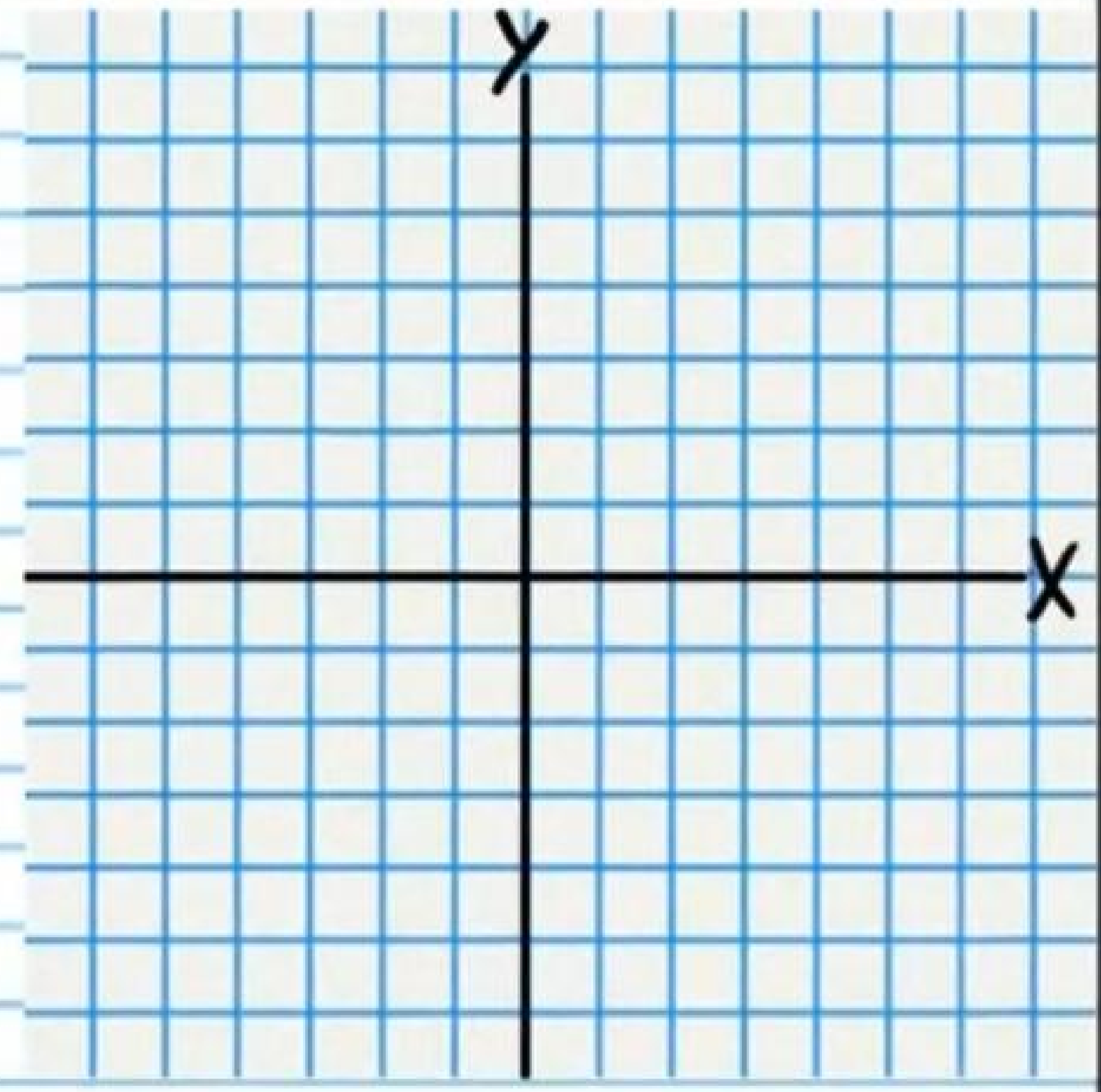
For each problem below, write an equation of the parabola with the given characteristics. Hint: It is easiest to use vertex form. Assume that all corresponding parabolas open up or down.

⑥ Vertex $(-1,-1)$ and directrix $y=-1\frac{1}{3}$

⑦ Vertex (2,5) and point (4,3).

⑧ Graph the following quadratic expression. Write the coordinates of the vertex and the x-intercept(s).

$y=\frac{1}{3}(x-1)^2-3$



⑨ Find the coordinates of the focus of the parabola in problem #8, and draw the point on the graph.

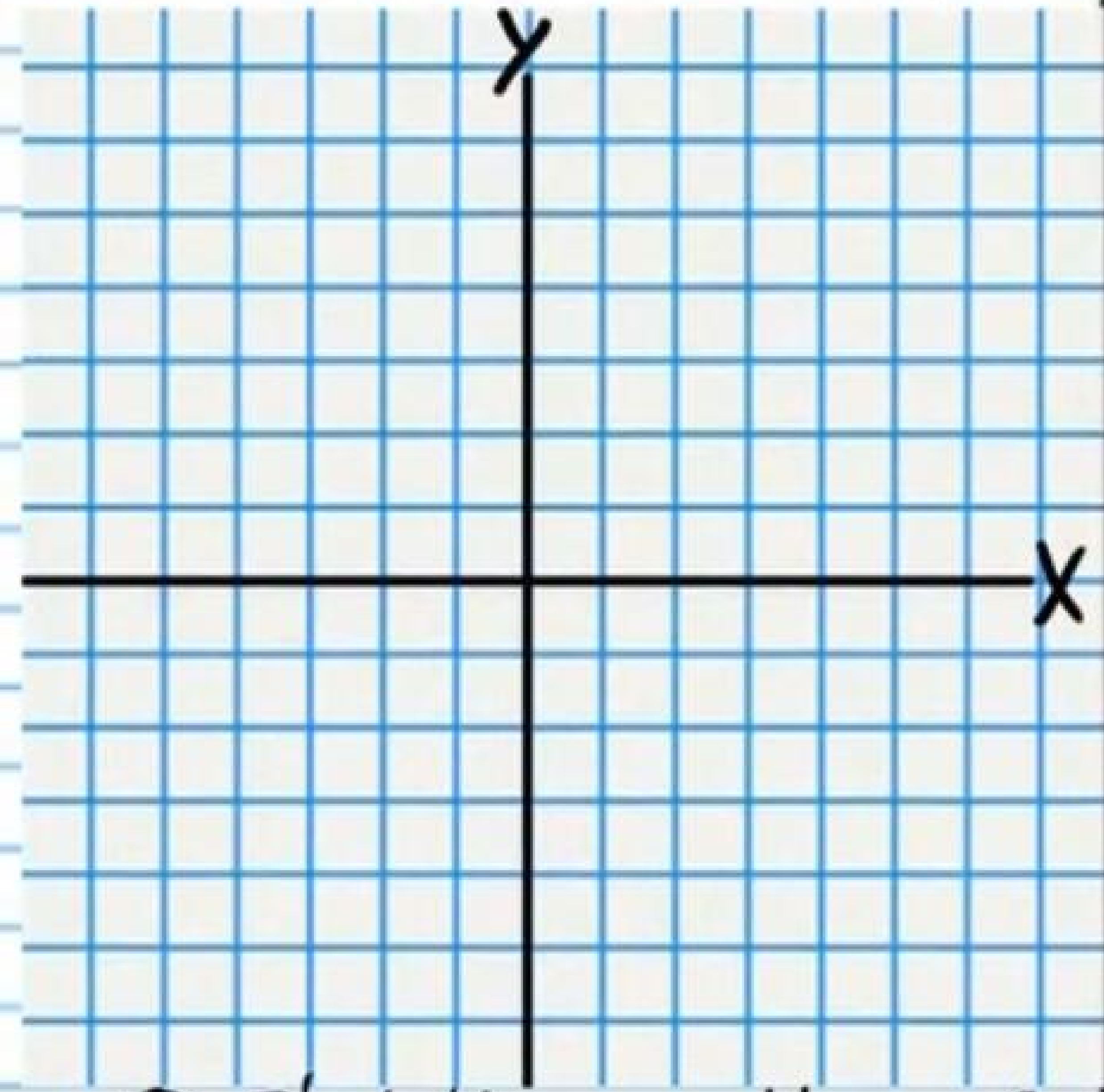
⑩ Find the equations of the line of symmetry and the directrix of the parabola in problem #8 and draw them (as dotted lines) on the graph.

(11) Find the coordinates of the x-intercepts of the parabola in problem #8.

(12) Find the y-intercept of the parabola in problem #8.

(13) Graph the following quadratic expression. Write the coordinates of the vertex and the x-intercept(s) on the graph.

$$\frac{1}{8}(x-2)(x+2)=y$$

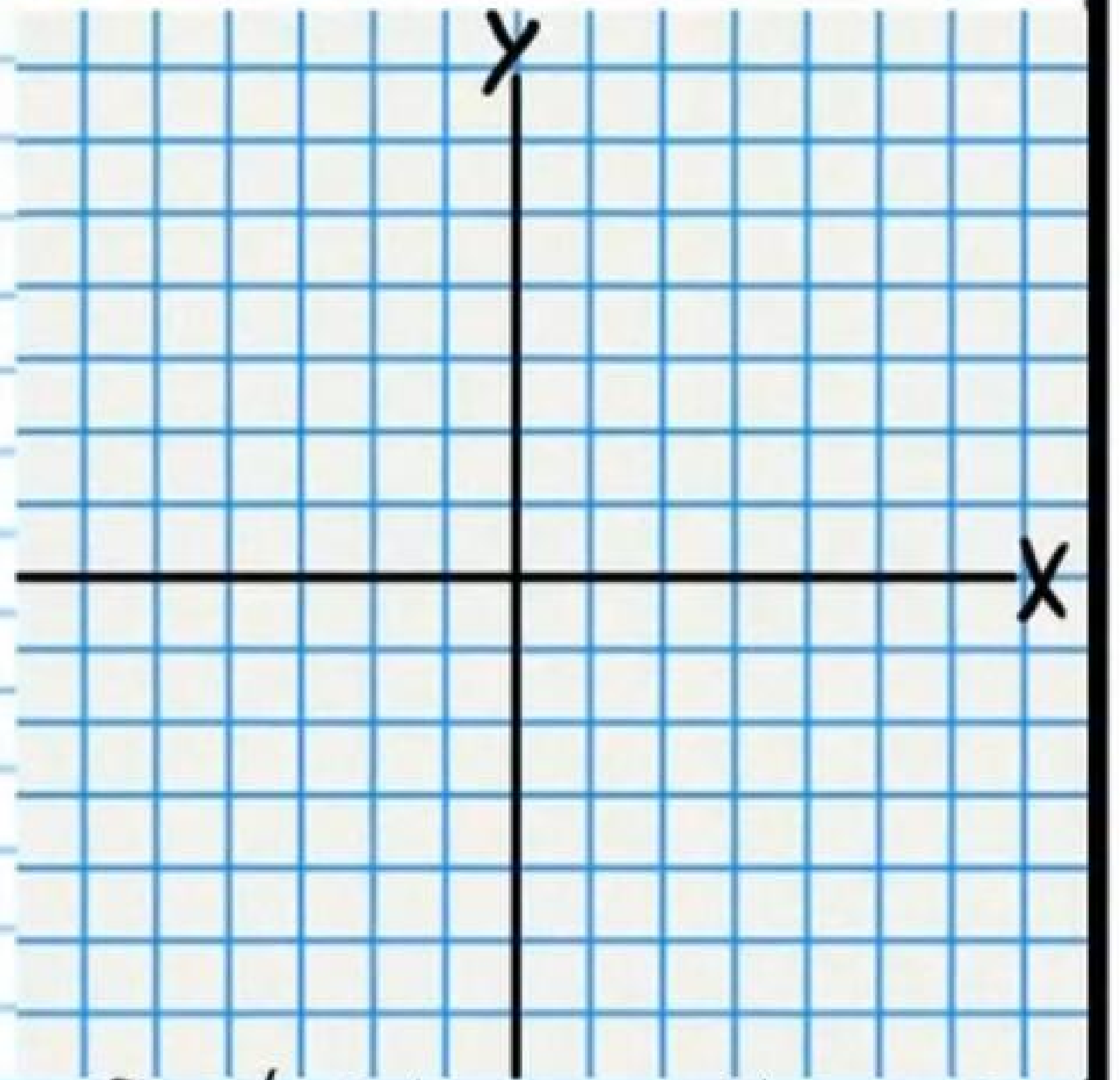


(14) Find the coordinates of the focus of the parabola in problem #13, and draw the point on the graph.

(15) Find the equations of the line of symmetry and the directrix of the parabola in problem #13, and draw them (as dotted lines) on the graph.

(16) Graph the following quadratic expression. Write the coordinates of the vertex and the x-intercept(s).

$y = -x^2 - 2x$

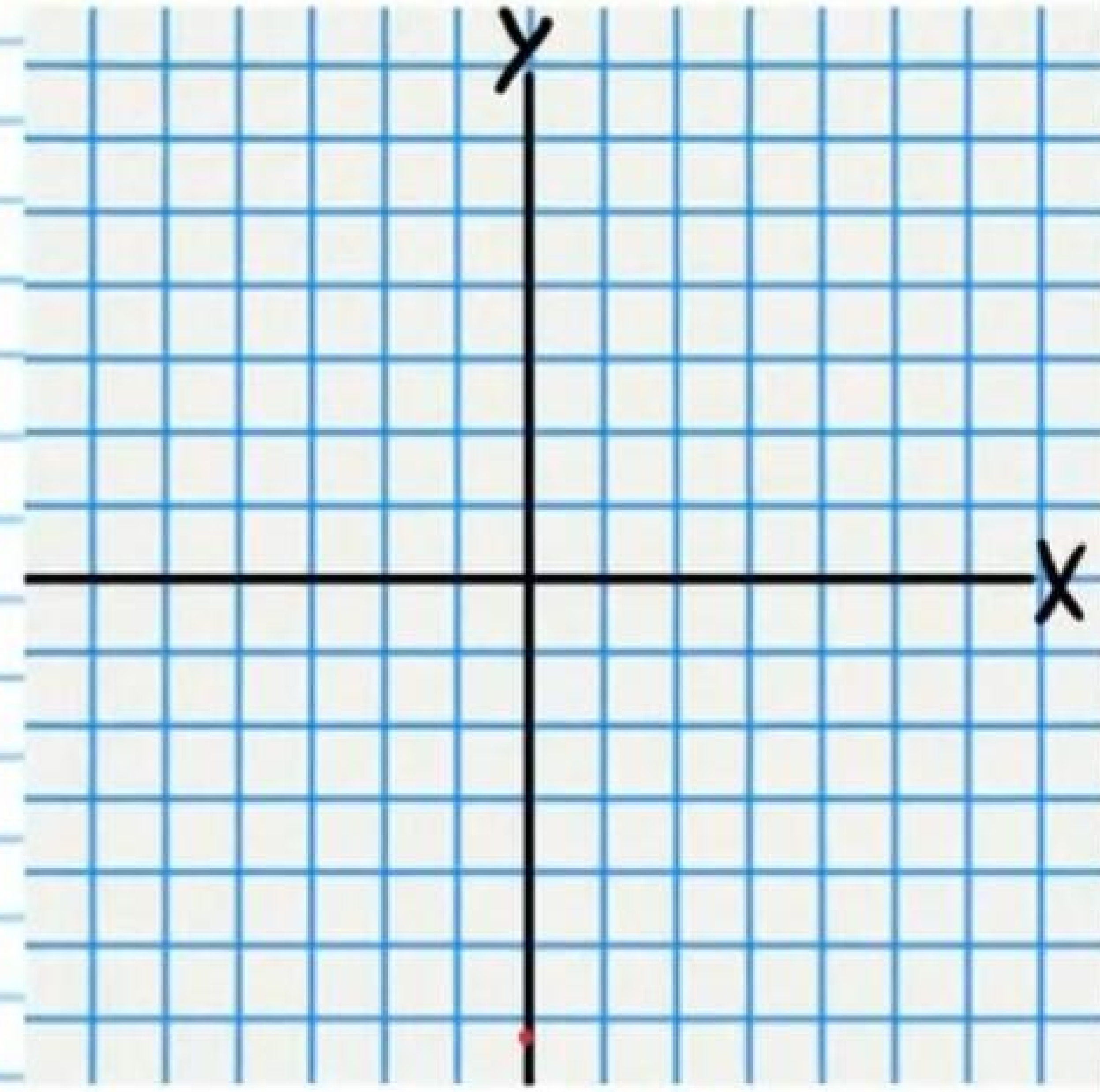


(17) Find the coordinates of the focus of the parabola in problem #16, and draw the point on the graph.

(18) Find the equations of the line of symmetry and the directrix of the parabola in problem #16, and draw them (as dotted lines) on the graph.

(19) Graph the following quadratic expression. Write the coordinates of the vertex and the x-intercept(s).

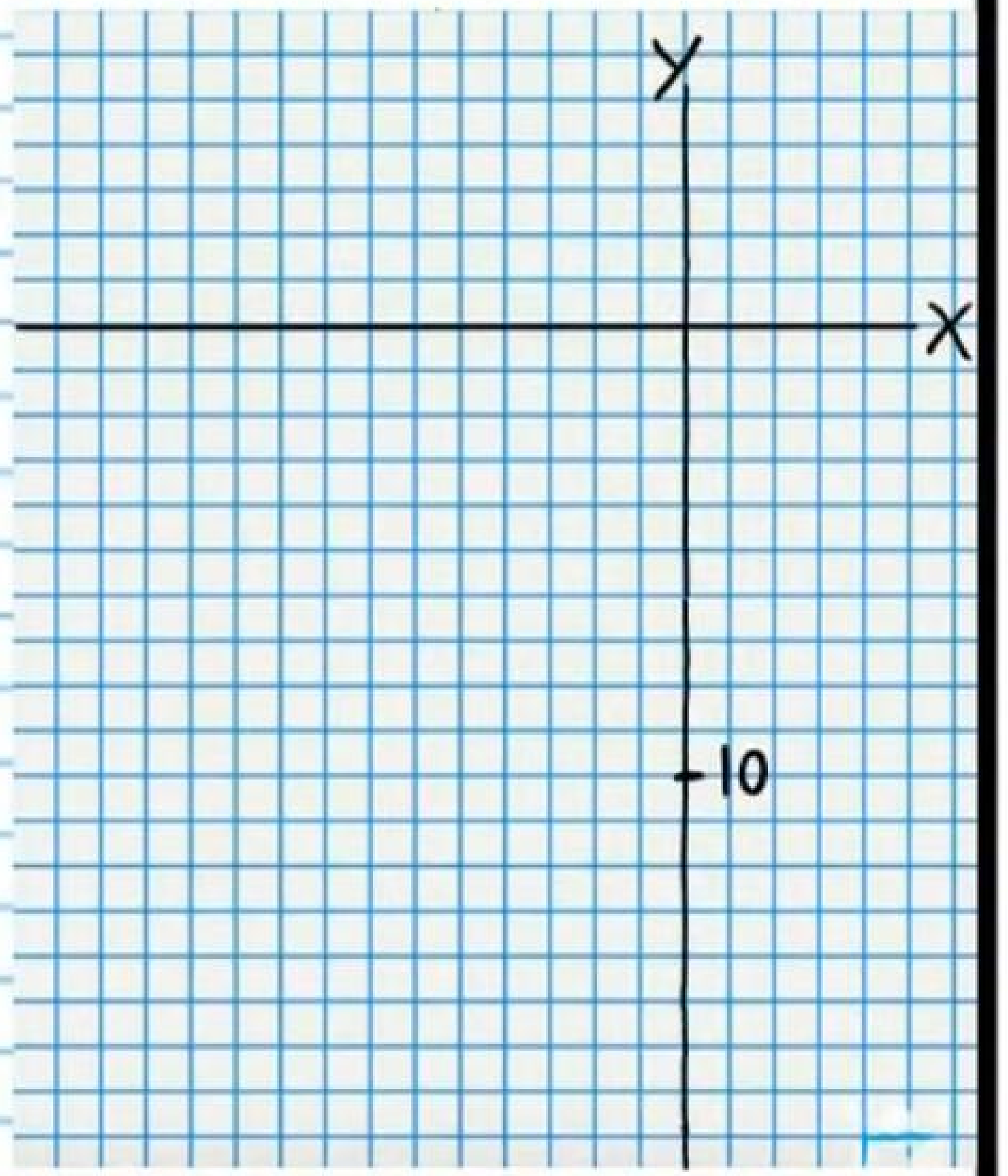
$-\left(x-\frac{5}{2}\right)\left(x-\frac{5}{2}\right) = y$



(20) Find the y-intercept of the parabola in problem #19.

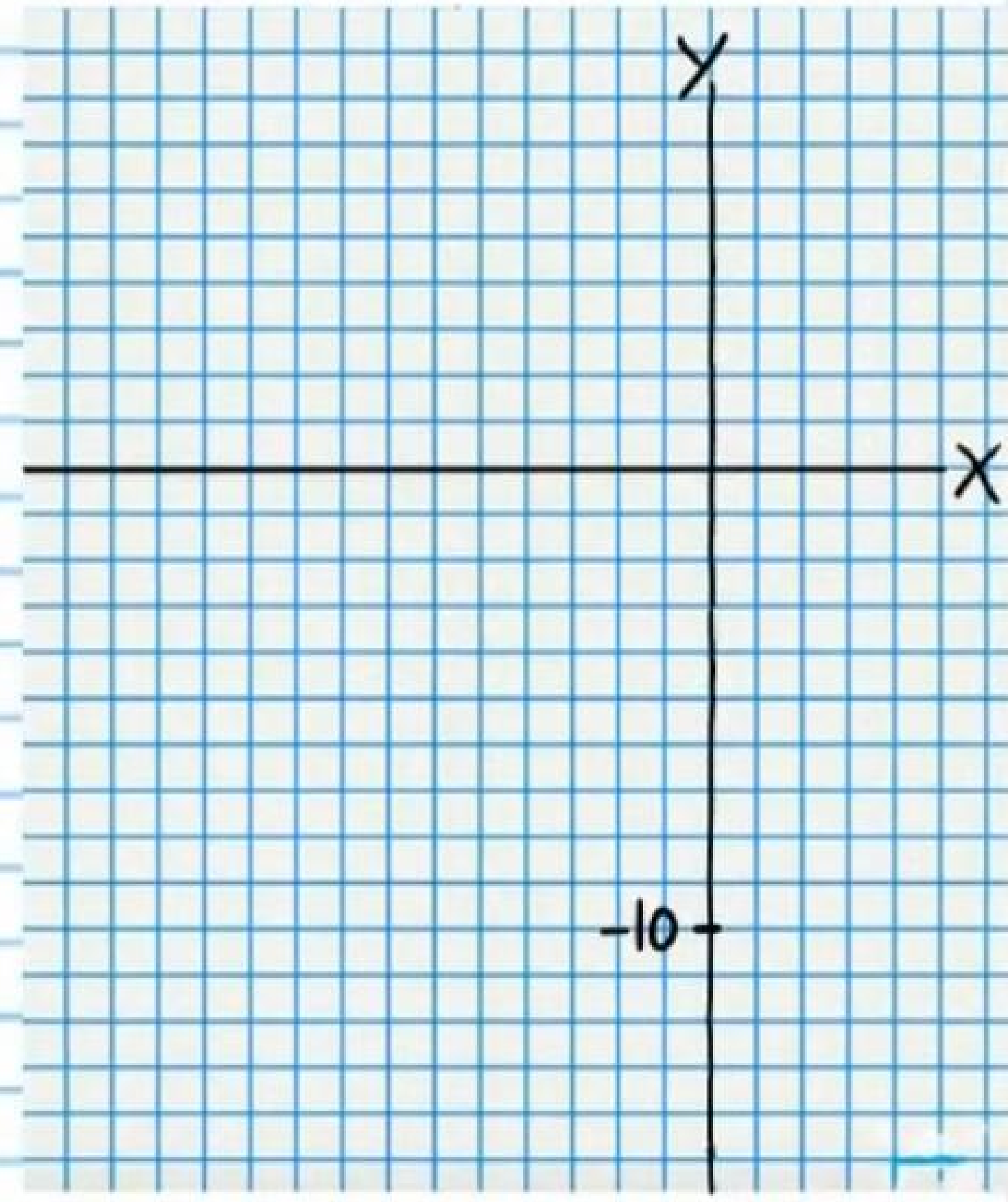
21. Graph the following quadratic expression. Write the coordinates of the vertex, the x-intercept(s), and the y-intercept.

$y = -2(x+3)^2$



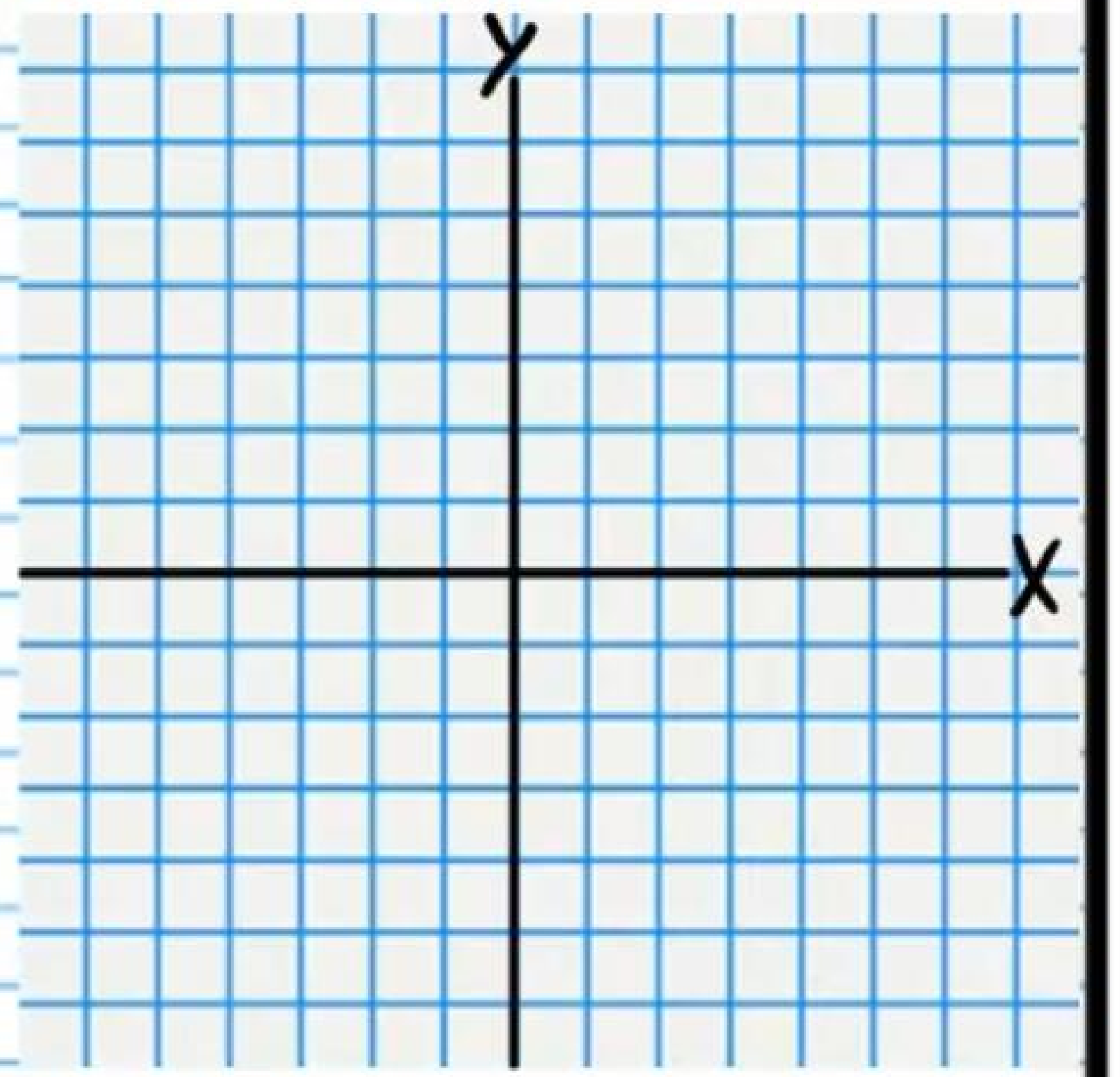
22. Graph the following quadratic expression. Write the coordinates of the vertex and the x-intercept(s).

$x^2 + 3x - 10 = y$



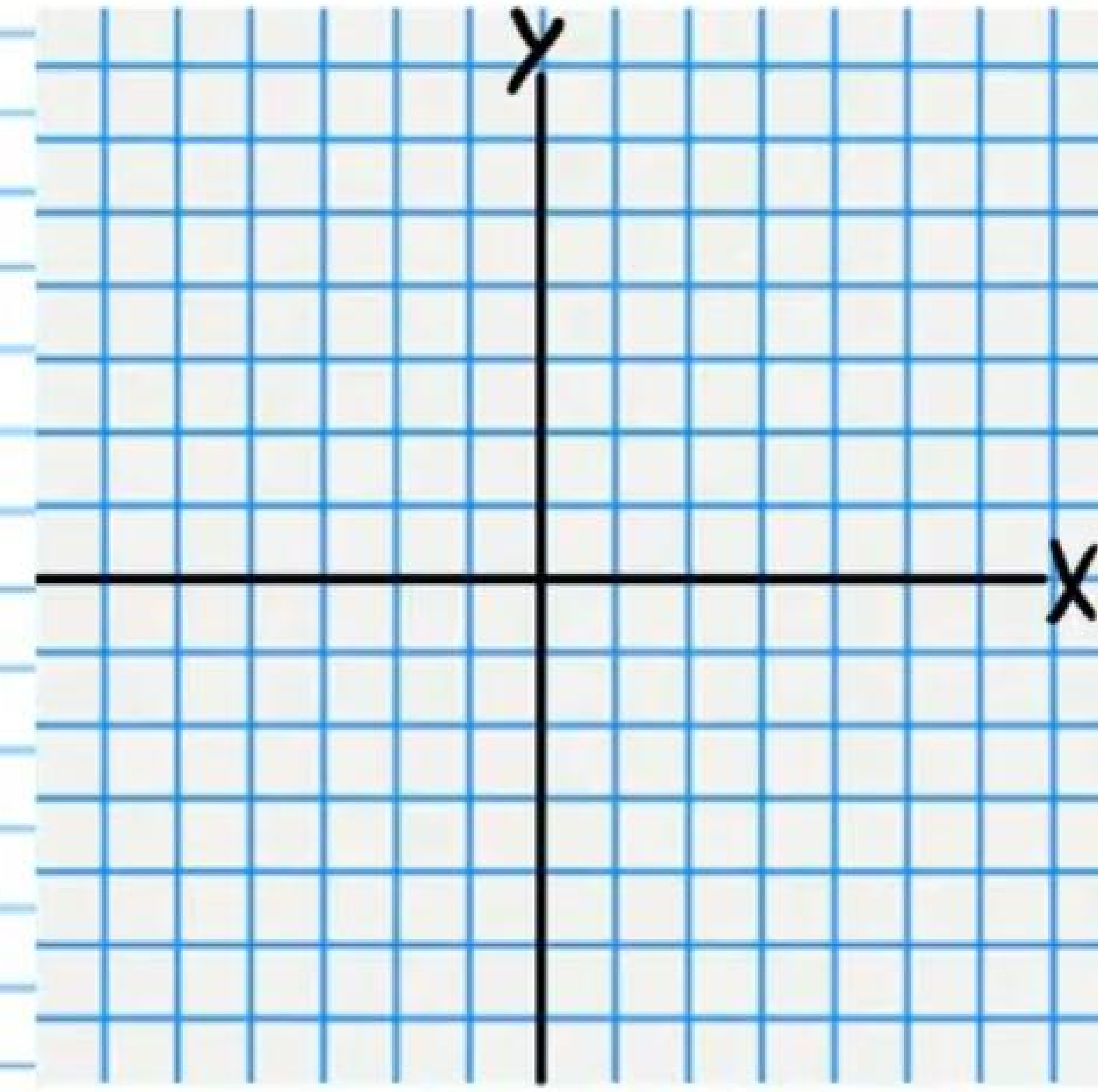
(23) Graph the following quadratic expression. Write the coordinates of the vertex and the x-intercept(s).
Write exact expressions for the x-intercepts.

$y = 3x^2 + 12x + 11$



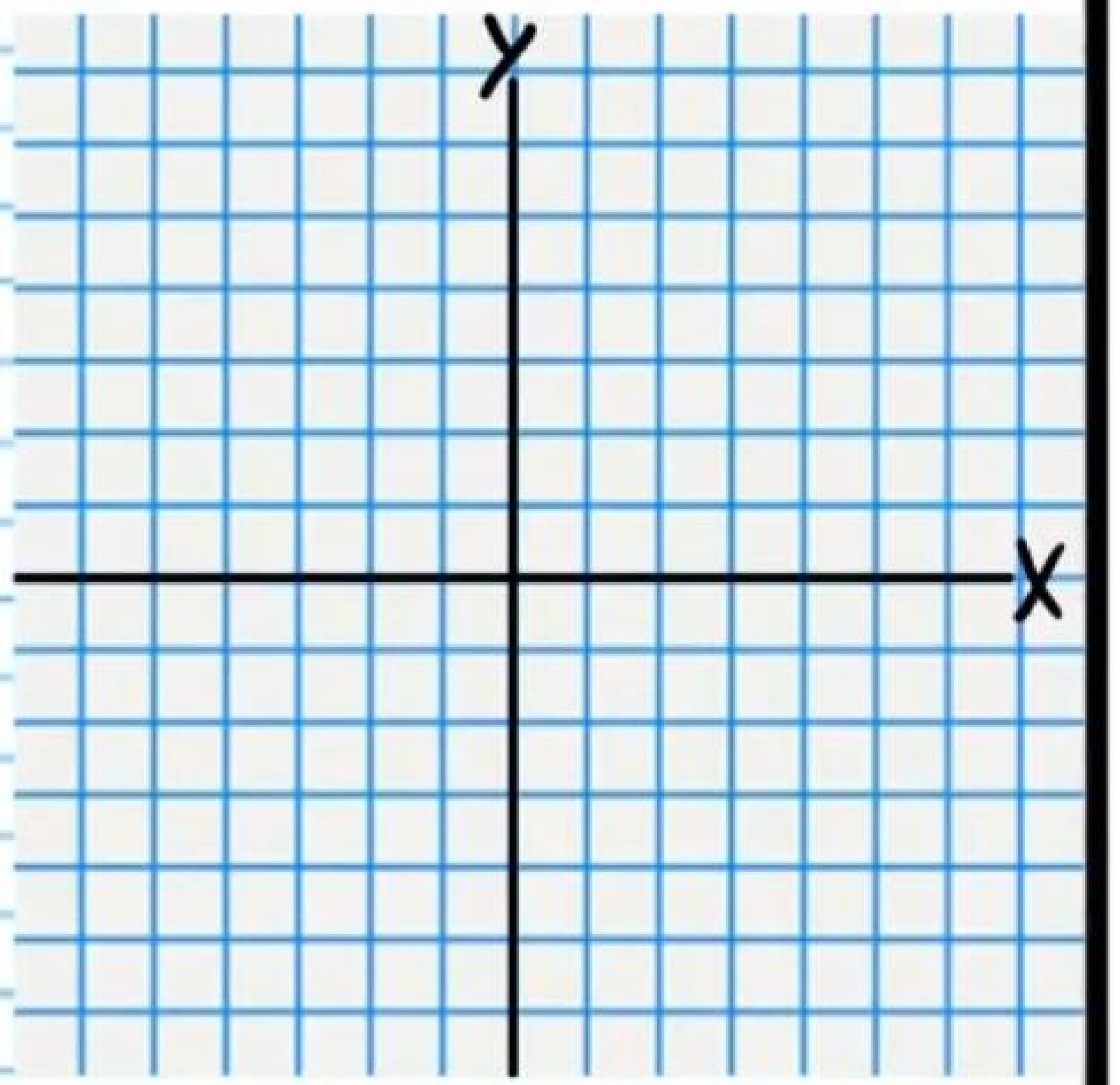
(24) Graph the following quadratic expression. Write the coordinates of the vertex and the x-intercept(s).
Write exact expressions for the x-intercepts.

$-(x-3)^2 + 6 = y$



(25) Graph the following quadratic expression. Write the coordinates of the vertex, the x-intercept(s), and the y-intercept.

$y = x^2 + 2x + 2$



Answers

① $\boxed{y=(x+5)^2-4}$

② $\boxed{3(x-1)(x+3)=y}$

③ $\boxed{y=5(x+3)^2-20}$

$\boxed{y=5(x+1)(x+5)}$

④ $\boxed{D=5}$

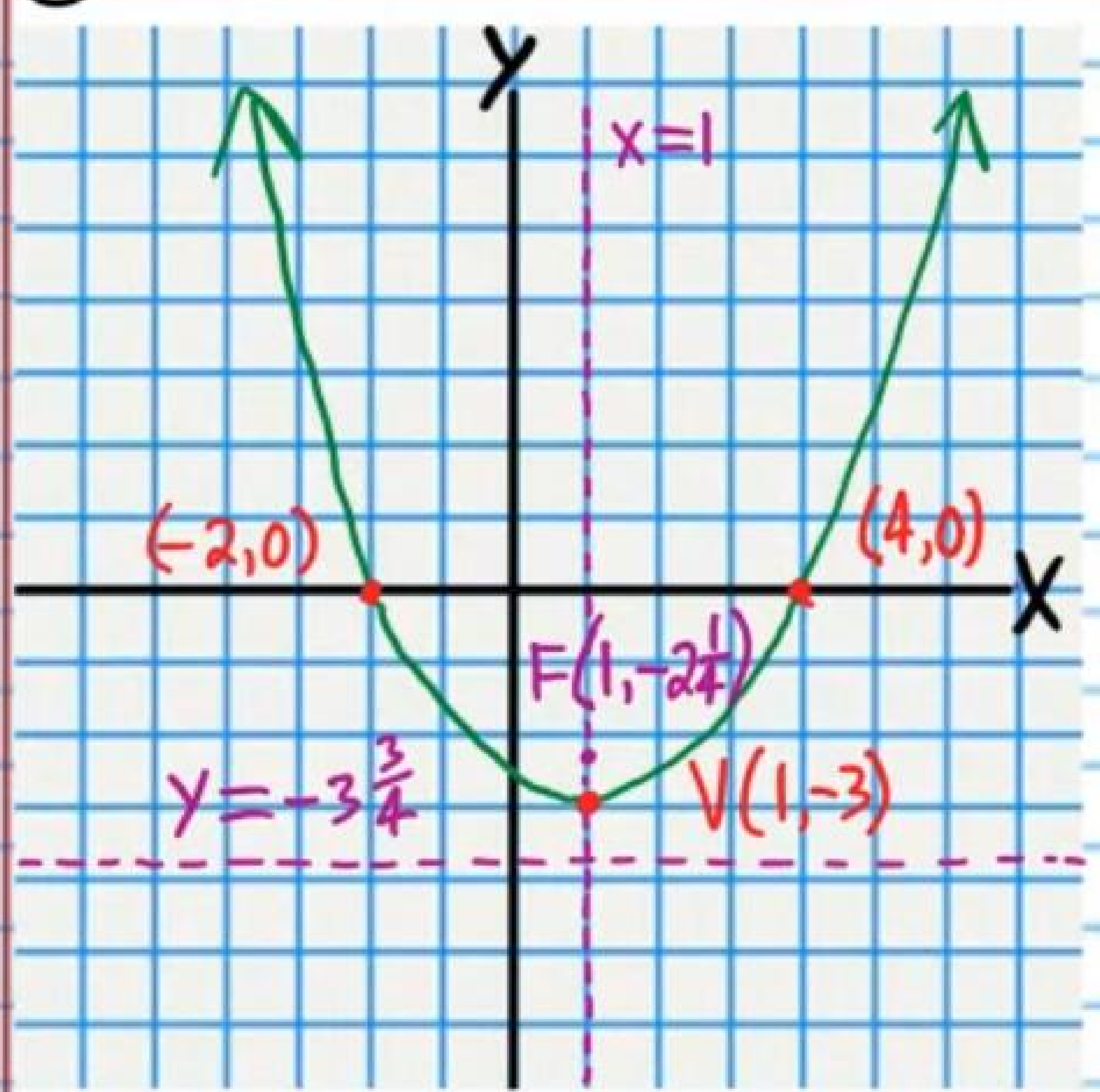
$\boxed{\text{Two x-intercepts}}$

⑤ $\boxed{y=-4(x-3)^2+6}$

⑥ $\boxed{y=\frac{3}{4}(x+1)^2-1}$

⑦ $\boxed{y=-\frac{1}{2}(x-2)^2+5}$

⑧



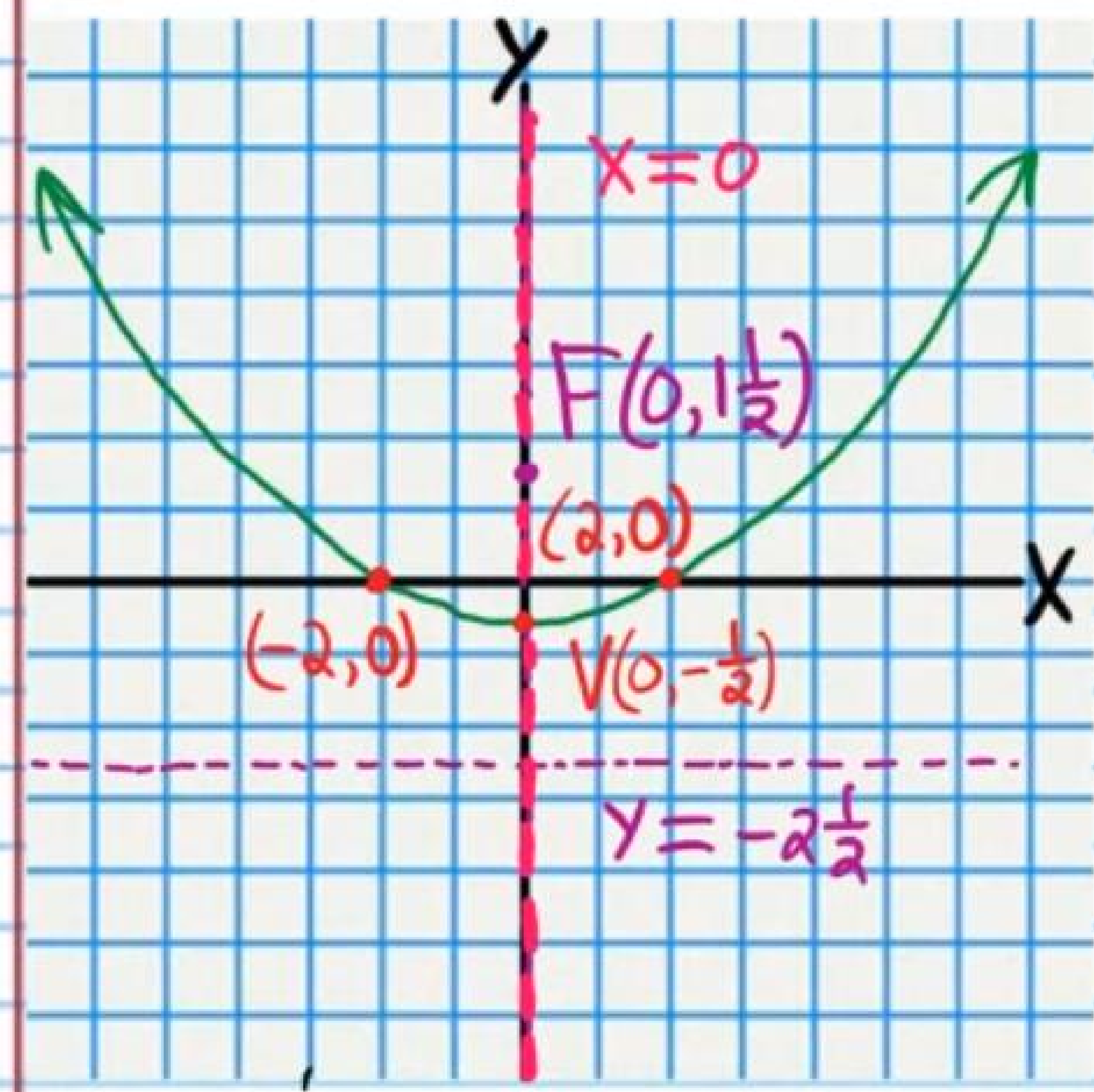
⑨ $\boxed{F(1,-2\tfrac{1}{4})}$

⑩ $\boxed{x=1}$ $\boxed{y=-3\tfrac{3}{4} \text{ or } -\tfrac{15}{4}}$

⑪ $\boxed{(-2,0) \ \& \ (4,0)}$

⑫ $\boxed{-2\tfrac{2}{3}}$ or $\boxed{(0,-2\tfrac{2}{3})}$

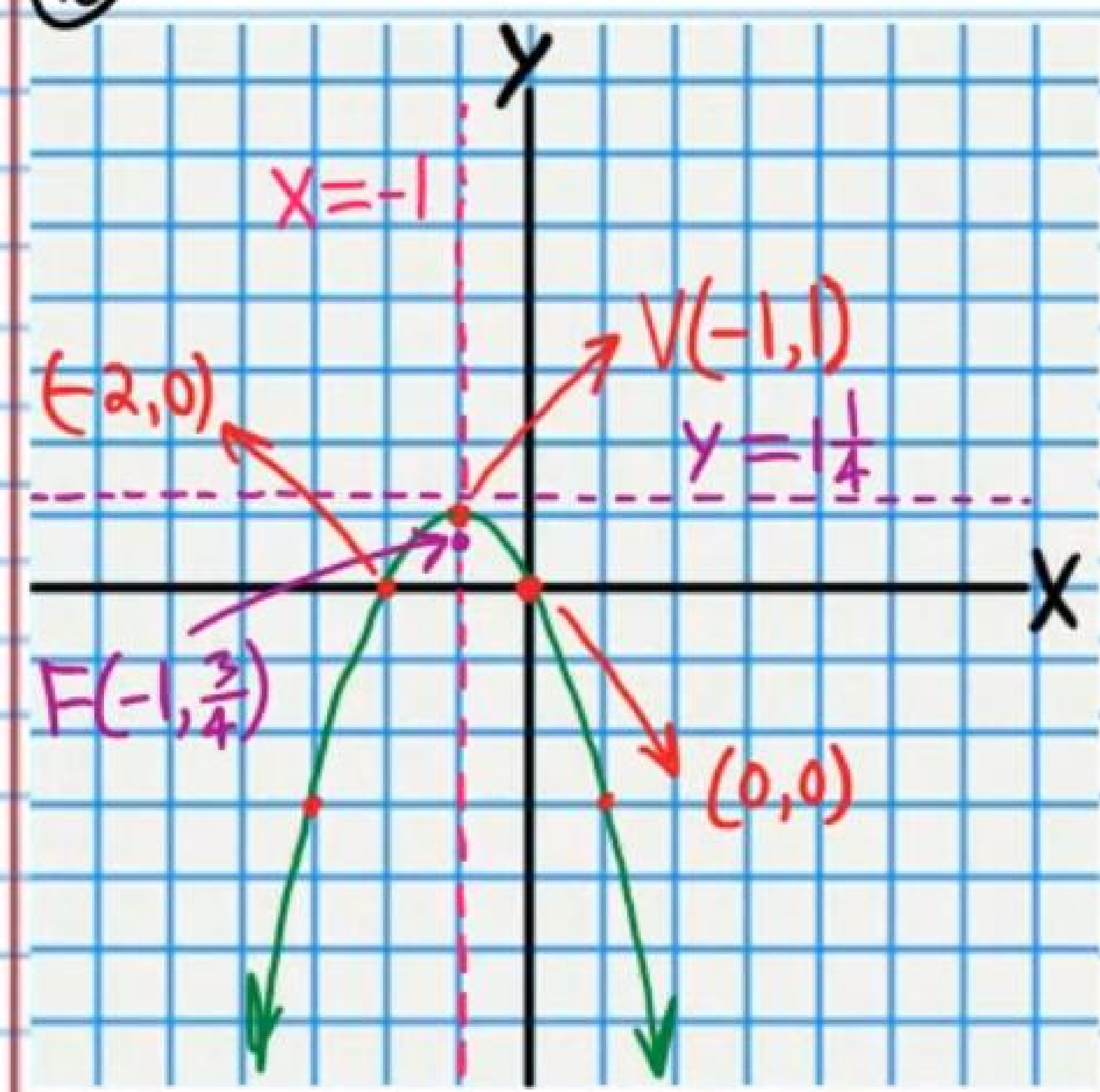
⑬



⑭ $\boxed{F(0,1\tfrac{1}{2})}$

⑮ $\boxed{x=0}$ $\boxed{y=-2\tfrac{1}{2}}$

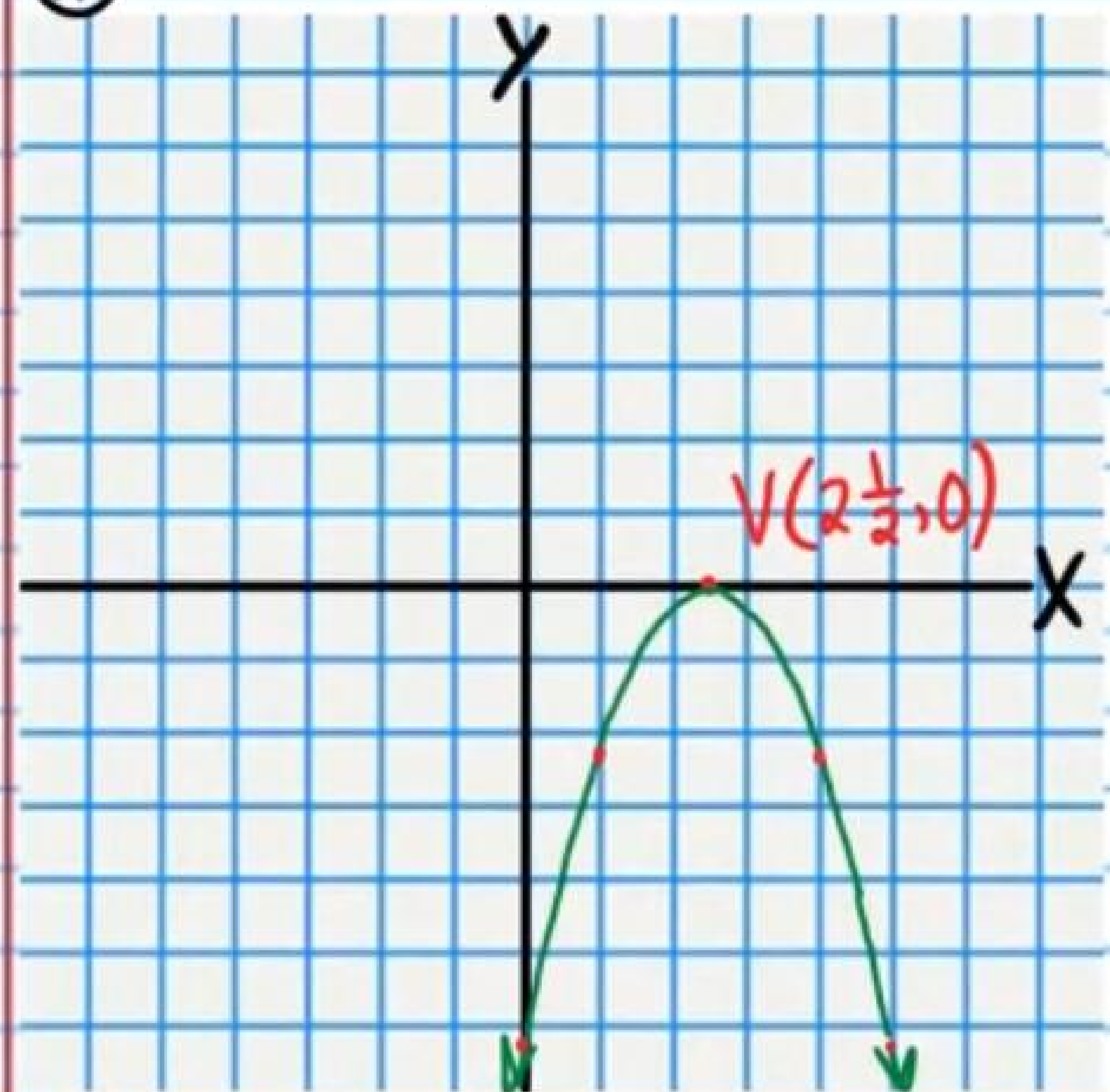
⑯



⑰ $\boxed{F(-1,\tfrac{3}{4})}$

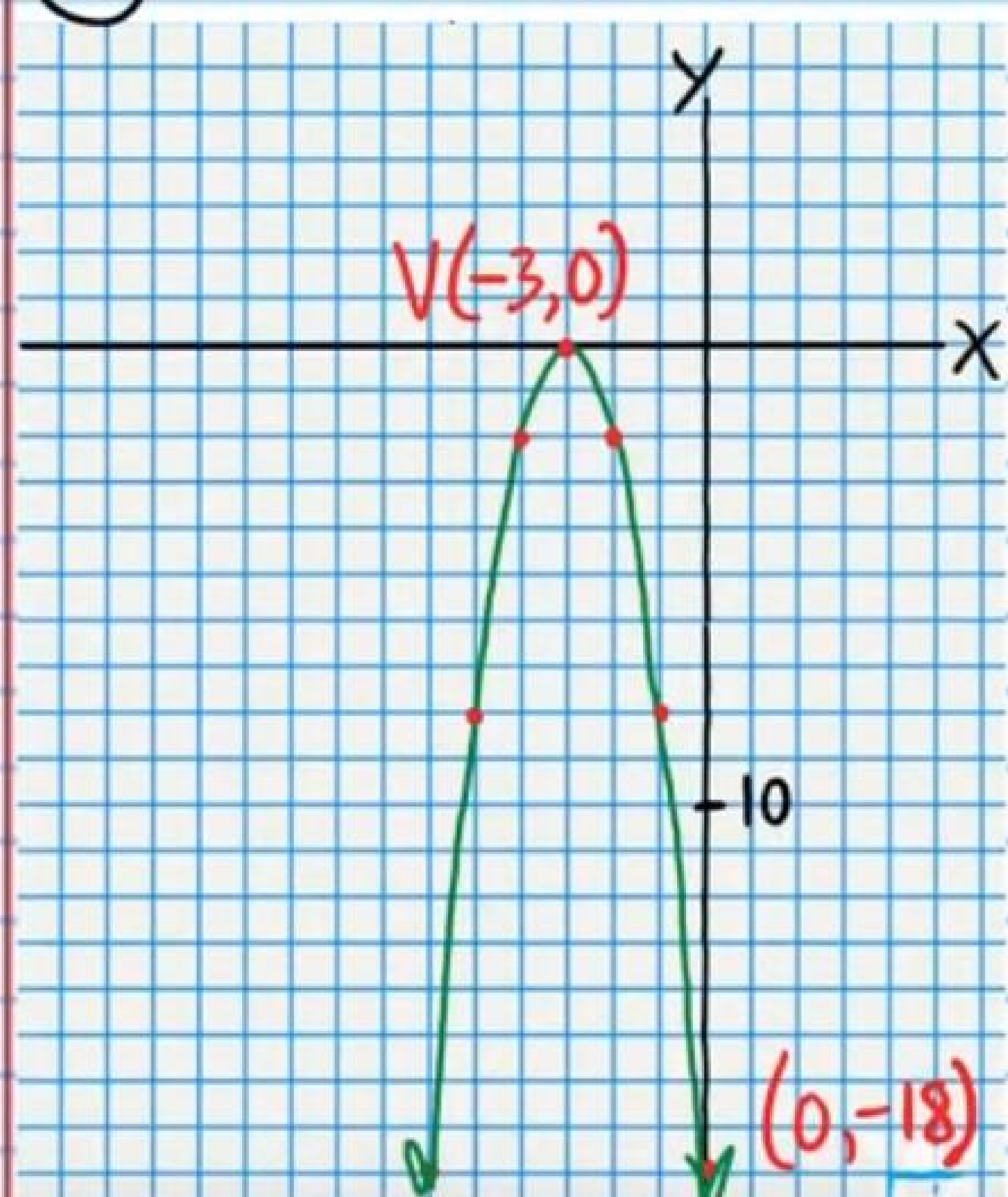
⑱ $\boxed{x=-1}$ $\boxed{y=1\tfrac{1}{4} \text{ or } \tfrac{5}{4}}$

19

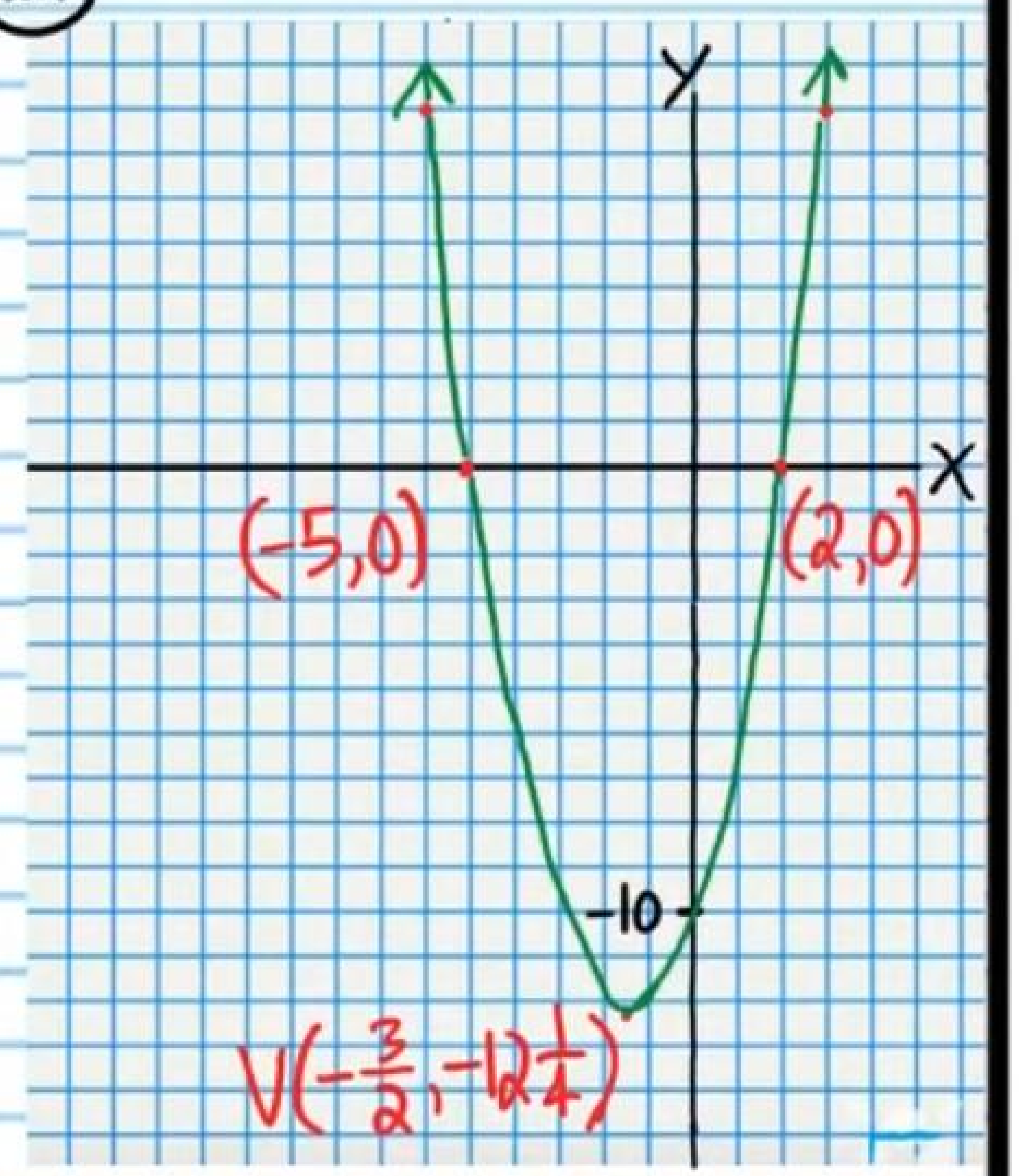


20 $\boxed{(0,-6\frac{1}{4})}$

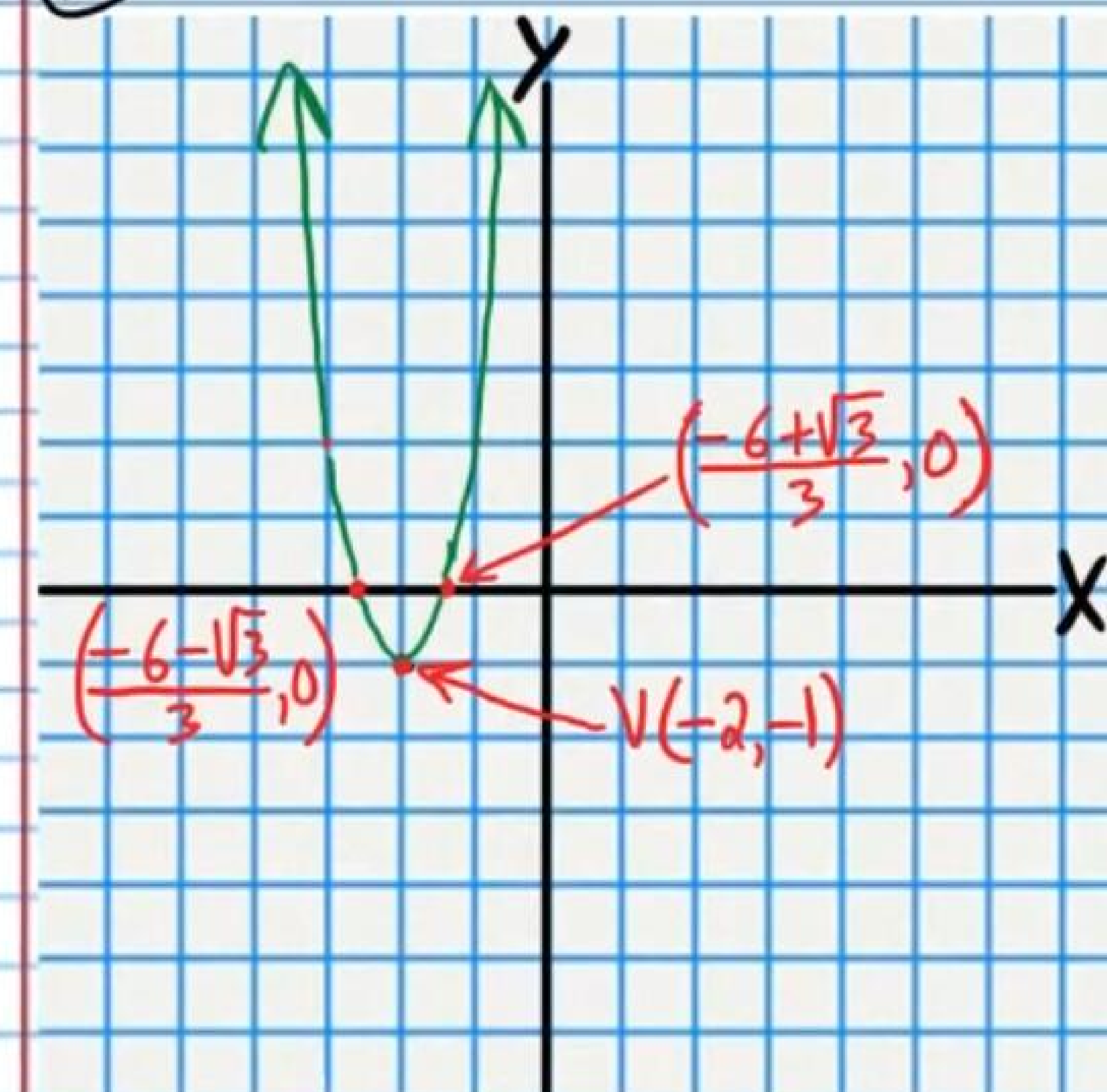
21



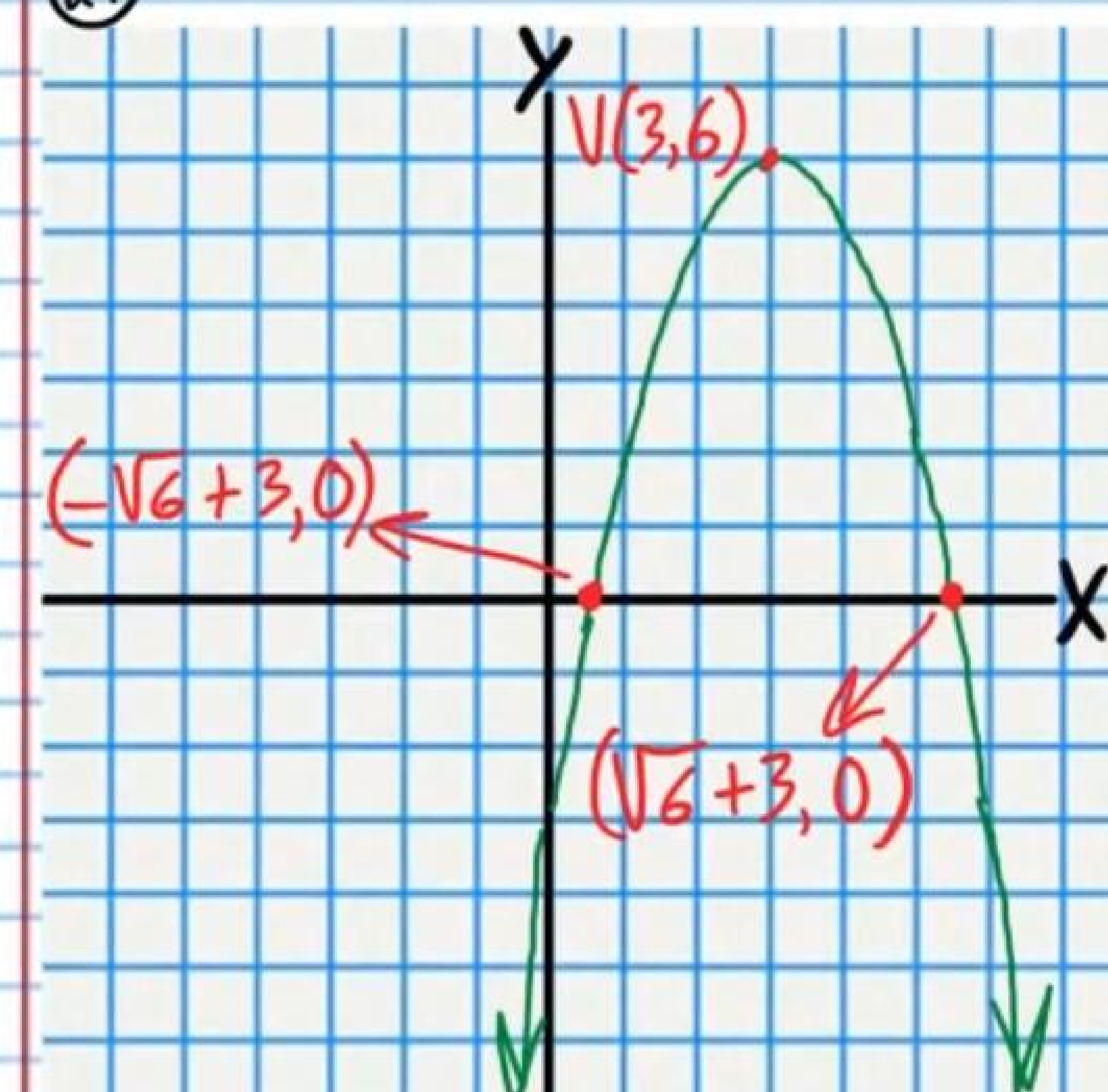
22



23



24



25